# Co-speech gestures modulate metaphor comprehension, but listeners still favor default interpretations.

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**Introduction.** Previous research has shown that gesture and speech are automatically integrated during sentence comprehension, with most studies measuring the effect of incongruent gesture–speech pairings on response times [1] or electrical brain responses [2] compared to congruent ones. But how does sentence comprehension change when co-speech gestures support different *congruent* interpretations of the same expression? In this study, we employed 24 English metaphors, such as *Men are oaks*, with varying degrees of familiarity. These expressions afford a preferred interpretation (e.g., robust), a possible but less salient interpretation (e.g., tall), and an impossible interpretation (e.g., hikers). We manipulated both interpretation type (preferred, possible, impossible) and gesture condition (gesture-of-preferred, gesture-of-possible, and non-gestured) to examine the extent to which gesture can modulate—or even shift—listeners' default interpretation. In Experiment 1, we examined whether the perceived meaningfulness of preferred, possible, and impossible interpretations was influenced by co-speech gestures matching either the preferred interpretation, the possible interpretation, or by the absence of gesture when participants were able to rate all interpretations (see Fig. 1). In Experiment 2, we examined whether this effect changes when participants are required to choose a single interpretation. **Method.** Participants watched short video clips of a speaker producing an expression with the three interpretations displayed below the video. Gesture condition was a within-subjects factor, and participants were not informed that some videos contained gestures while others did not. This allowed us to examine whether gestures naturally influenced responses. In Experiment 1 (N = 60, age range: 21–70 years; M = 44.78, SD = 12.71; 25 female), the task was to rate how well each interpretation conveyed the meaning of the expression on a scale from 1 (not at all) to 5 (very much). In Experiment 2 (N = 60, age range: 25–65 years; M = 46.00, SD = 9.46; 33 female), the task was to choose one of the three interpretations as the meaning conveyed by the speaker. **Results.** In Experiment 1, across all conditions, the preferred interpretation received higher meaningfulness ratings than both the possible and impossible interpretations, although this difference was smaller in the gesture-of-possible condition (see Fig. 2). However, meaningfulness ratings for the possible interpretation were higher in the gesture-of-possible condition compared to both the gesture-of-preferred condition ($\beta = 1.32$, 95% CRI [0.99, 1.65]) and the non-gestured condition ($\beta = 0.94$, 95% CRI [0.62, 1.27]). Moreover, meaningfulness ratings for the preferred interpretation were lower in the gesture-of-possible than in the gesture-of-possible preferred ($\beta = -1.34$, 95% CRI [−1.67, -1.01]). In Experiment 2, across all conditions, the preferred interpretation also showed higher probabilities of being selected than both the possible and impossible interpretations (see Fig. 3). However, the probability of selecting the possible interpretation increased in the gesture-of-possible condition compared to both the gesture-of-preferred condition ($\beta = 1.95$, 95% CRI [1.58, 2.33]) and the non-gestured condition ($\beta = 1.42$, 95% CRI [1.04, 1.77]). Moreover, the probability of selecting the preferred interpretation was lower in the gesture-of-possible condition compared to the gesture-of-preferred condition ($\beta = -1.87$, 95% CRI [−2.24, -1.51]). Moreover, in the gesture-of-possible condition, the mean posterior probability of choosing the preferred, possible, and impossible interpretations were 0.56, 0.41, and 0.02, respectively . **Discussion.** These results indicate that co-speech gestures influence metaphor comprehension, yet listeners still favor the default interpretation. When participants were allowed to consider and rate both the preferred and possible interpretations, both received relatively high ratings. However, when participants were required to choose a single interpretation, they typically chose the default one—even when the speaker gestured an alternative possible interpretation.

(a) robust (b) tall (c) non-gestured

**Figure 1.** Video frames showing gestures for the metaphor *Men are oaks,* corresponding to the two plausible interpretations of the expression and a non-gestured condition.

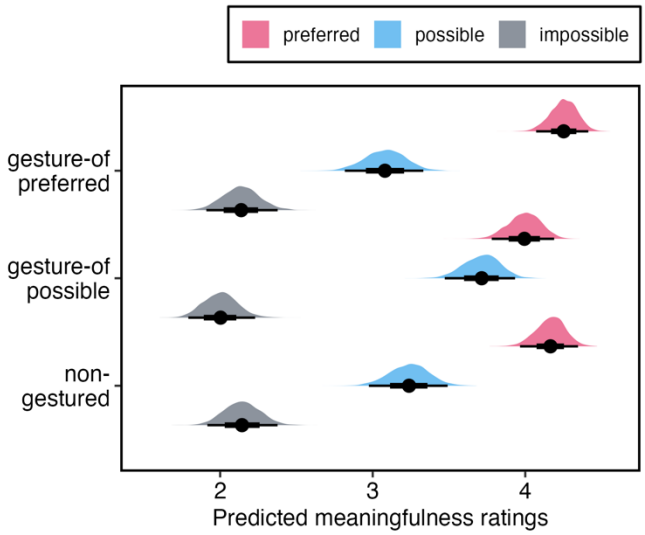


**Figure 2.** Predicted meaningfulness ratings for the preferred, possible, and impossible interpretations as a function of gesture condition in Experiment 1.

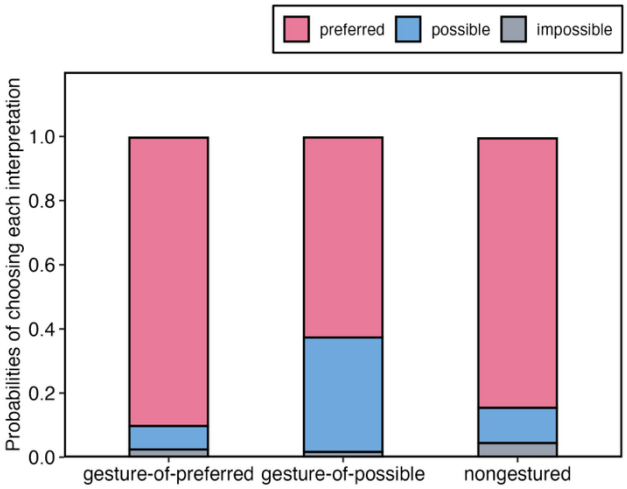


**Figure 3.** Predicted probability of selecting the preferred, possible, or impossible interpretations as a function of gesture condition in Experiment 2.

## References


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